Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/28/2026
PI Name:	Sackett, Charles Ph.D.		
Project Title:	Development of Atom Interferometry Experiments for the International Space Station's Cold Atom Laboratory		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	22904-1000	**Congressional District:**	5
Comments:	Other names: CA Sackett; Cass Sackett.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
Start Date:	04/01/2014	**End Date:**	03/20/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 03/20/2026 per J. Williams/JPL (Ed., 2/4/26). NOTE: End date changed to 9/30/2025 per D. Griffin/NASA-HQ (Ed., 1/20/25). NOTE: Contract number changed to JPL 1700183 per J. Callas/JPL (Ed., 8/29/24). NOTE: End date changed to 9/27/2024 per U. Israelsson/JPL (Ed., 1/6/22) NOTE: Extended to 9/30/2022 per U. Israelsson/JPL (Ed., 3/9/21) NOTE: Extended to 10/28/2020 per PI (Ed., 2/28/2020) NOTE: Extended to 10/30/2019 per U. Israelsson/JPL (Ed., 12/14/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1700183 ; JPL 1502012		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The ultimate objective of this proposal is to develop an ultra-high sensitivity atom interferometer capable of operating in and benefiting from a microgravity environment. The interferometer would be specifically suited for measurements of rotations, but it would be broadly applicable to a variety of precision measurements. Ground and flight based efforts are proceeding in three broad areas. First, we are performing ground studies and developing a flight mission for the Cold Atom Laboratory (CAL) to study atomic techniques for inertial sensing in microgravity. Ground efforts include development of new rotation-sensing techniques and implementation of an optically suspended atom source for gravimetry. Flight efforts involve implementation and characterization of atom interferometry techniques using the CAL apparatus on the International Space Station (ISS). Second, we are investigating methods to produce an ultra-low temperature atom source in free space using the CAL apparatus. The apparatus produces atoms confined in a magnetic trap, but inertial measurements require free atoms. We will investigate releasing the atoms by gradually turning off the trapping fields, allowing the atoms to adiabatically expand and cool off. This can produce a relatively dense and very low-velocity sample that is ideal for atom interferometry methods. Third, we will continue ground-based studies to develop novel precision measurement techniques for use with atom interferometry, such as tune-out spectroscopy. Techniques like this are useful for advancing scientific knowledge and would be good candidates for future flight studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of precision inertial sensing techniques is useful for Earth-based as well as space-based navigation. Besides using direct sensing for inertial navigation, rotation sensing can also be useful for north-finding while gravity sensing can be used to tabulate local gravity variations and form a type of three-dimensional map for navigating. These techniques also have many applications in geophysics. Gravity sensing can be used for oil and mineral exploration, while rotation sensing can detect dynamics in the Earth's core. Gravity sensing also has defense applications such as locating underground tunnels and potential screening cargo for high-density contraband or weapons. Other precision measurement applications have less direct impact, but advance scientific knowledge. For instance, precision tune-out spectroscopy measurements of atomic matrix elements can be used to improve the interpretation of atomic parity violation experiments. These in turn impact our understanding of the standard model of particle physics and thus the nature of our universe. Direct benefits of such understanding can be hard to trace, but in general the continued advance of technological applications builds on advances in our fundamental knowledge.
Task Progress:	A primary focus of the project is developing and carrying out flight experiments on the Cold Atom Lab (CAL) apparatus. Due to technical problems, CAL's capabilities were limited during most of the reporting period, such that atom interferometry was unavailable and productions of Bose-Einstein condensates was limited. In response to these constraints, we developed a project to investigate the behavior of non-condensed atoms in a magnetic quadrupole trap. CAL uses a quadrupole trap to help load its atom chip, and any improvements in loading efficiency can have an outsize impact on the number of condensed atoms eventually produced. A quadrupole magnetic trap is very simple, and is made using two electromagnet coils with opposing currents, such that the magnetic field is zero at the center. In CAL, laser cooled atoms from the magneto-optical trap are loaded into a quadrupole trap and then moved close to the chip surface. A quadrupole trap can provide very tight confinement, but it is unsuitable for producing Bose condensation because atoms with sufficiently low energy spend most of their time near the center of the trap where the magnetic field is zero. In a sufficiently low field, a moving atom can undergo a so-called Majorana transition in which it changes its spin quantum state. After undergoing this transition, the interaction with the trapping field is inverted such that the atom is expelled from the trap. This is a form of loss that limits the lifetime of atoms in the trap, with colder atoms having a shorter lifetime. Our goal in these experiments is to measure the Majorana loss rate as a function of the atom temperature and trap strength. This can be compared to ground measurements, where gravity plays a significant role in weaker traps. We have also measured the atom number and temperature, from which the collision rate between atoms can be calculated. If the collision rate is sufficiently large compared to the loss rate, it should be possible to perform a limited state of evaporative cooling in the quadrupole trap. This would reduce the atom temperature without reducing the phase space density, which would improve the efficiency of transfer atoms to the atom chip trap. Measurements so far have given lifetimes consistent with Majorana expectations, and collision rates high enough that evaporative cooling seems feasible. A challenge for both the measurements and the evaporative cooling idea is that the CAL magnetic coils can only be operated for a limited time to avoid overheating. We circumvent this constraint for the lifetime measurements by adiabatically expanding the trap, yielding colder atoms and requiring lower coil currents. For the evaporative cooling, we will explore how quickly the cooling can be implemented while still observing some benefit in atom temperature and phase space density. We are also carrying out ground experiments in our lab at the University of Virginia, focused on development of an atom gyroscope for inertial navigation. During the reporting period, we have rebuilt the apparatus to work with atom chips, similar to how CAL works. In our case, we use a high power laser to produce "optical tweezers" which transport atoms to the chip. We have successfully loaded atoms into the tweezers and transported them, but we uncovered a flaw in the vacuum chamber design which prevented us from moving the atoms all the way to the chip. We have now repaired this flaw and are bringing the apparatus back online.
Bibliography Type:	Description: (Last Updated: 04/22/2026)
Abstracts for Journals and Proceedings	Sackett C. "Quantum sensing using trapped atom interferometry." Quantum Days 2025, Toronto, Canada, February 19-21, 2025. Abstracts. Quantum Days 2025, Toronto, Canada, February 19-21, 2025. https://2025.quantumdays.ca/en/program/ , Feb-2025

Abstracts for Journals and Proceedings	Chu Z, Sackett C. "Quantum rotation sensing by BEC in an atom chip waveguide." 56th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics, Portland, Oregon, June 16-21, 2025. Abstracts. 56th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics, Portland, Oregon, June 16-21, 2025. Poster. https://archive.aps.org/damop/2025/d00/145/ , Jun-2025
Abstracts for Journals and Proceedings	Fowler NJ. "Noise sensitivity of differential interferometry." 92nd Annual Meeting of the Southeastern Section of the American Physical Society, Harrisonburg, Virginia, October 23-25, 2025. Bulletin of the American Physical Society. 2025 Oct 23; Abstract: D01.00015. Poster. https://meetings.aps.org/Meeting/SES25/Session/D01.15 , Oct-2025
Abstracts for Journals and Proceedings	De Urioste Terrazas I, Sackett CA, Chu Z. "Fabrication and modeling of a cross-trap atom chip for interferometric applications." 92nd Annual Meeting of the Southeastern Section of the American Physical Society, Harrisonburg, Virginia, October 23-25, 2025. Bulletin of the American Physical Society. 2025 Oct 23; Abstract: D01.00013. Poster. https://meetings.aps.org/Meeting/SES25/Session/D01.13 , Oct-2025
Papers from Meeting Proceedings	Beydler MM, Moan ER, Sackett CA. "Quantum sensing using trapped atom interferometry." SPIE Quantum West 2025 (International Society for Optics and Photonics), San Francisco, California, January 25-29, 2025. SPIE Quantum West 2025 Proceedings: Quantum sensing, imaging, and precision metrology III. 2025 Mar 19;13392:1339202. https://doi.org/10.1117/12.3053194 , Mar-2025